Amrita School of Computing

B.Tech CSE(AI)

22AIE305 Cloud Computing

Lab Sheet 3

Aim

To understand Amazon Elastic Compute Cloud (EC2) by launching a virtual machine, configuring security groups, connecting to the instance, installing an Apache Web Server, and hosting a simple HTML webpage. Learning IAAS.

Learning Outcomes

1. After completing this experiment, students will be able to:
2. Explain Infrastructure as a Service (IaaS).
3. Launch an EC2 instance using the AWS Free Tier.
4. Configure Security Groups for inbound traffic.
5. Connect to an EC2 instance using EC2 Instance Connect.
6. Install and configure the Apache Web Server.
7. Host a static HTML webpage on EC2.
8. Access the hosted webpage using the public IPv4 address.
9. Stop and terminate an EC2 instance to avoid unnecessary charges.

Amazon EC2

Amazon Elastic Compute Cloud (EC2) is a web service provided by AWS that allows users to create virtual servers in the cloud. These virtual servers are called instances. EC2 enables users to run applications without purchasing or maintaining physical hardware.

Unlike Amazon S3, which stores files and hosts only static websites, EC2 provides a complete virtual computer where users can install operating systems, software, databases, and web servers.

Amazon Machine Image (AMI)

An Amazon Machine Image (AMI) is a pre-configured template used to launch an EC2 instance. It contains the operating system and essential software.

Examples include:

- Ubuntu Server
- Amazon Linux
- Red Hat Enterprise Linux
- Windows Server

For this experiment, **Ubuntu Server** will be used.

Instance Type

An instance type specifies the virtual hardware allocated to an EC2 instance.

AWS Free Tier provides:

t2.micro or **t3.micro** (depending on the region)

These instance types include:

- 1 Virtual CPU
- 1 GB RAM

Security Groups

A Security Group acts as a virtual firewall that controls incoming and outgoing network traffic.

For this experiment, the following inbound rules are required:

Protocol	Port	Purpose
HTTP	80	Access the hosted webpage
SSH	22	Connect to the instance (used by EC2 Instance Connect)

Apache Web Server

Apache HTTP Server is one of the world's most widely used web servers. It delivers web pages to clients through HTTP.

The default website directory in Ubuntu is: /var/www/html

The default homepage is: index.html

Architecture

Student Browser



HTTP



Public IPv4 Address

▼

Amazon EC2 Instance

|

Apache Web Server

|

index.html

Experiment

Step 1: Login to AWS

1. Open https://console.aws.amazon.com
2. Sign in using your AWS account.

Step 2: Open EC2 Dashboard

1. Search for EC2 in the AWS search bar.
2. Select EC2 from the list of services.

Step 3: Launch an EC2 Instance

1. Click Launch Instance.
2. Enter the instance name:

   CloudLab-EC2

3. **Under Amazon Machine Image, select:**

   Ubuntu Server 24.04 LTS

4. **Under Instance Type, choose:**

   t2.micro (Free Tier Eligible)

Step 4: Configure Key Pair

Choose:

Proceed without a key pair

(We will use EC2 Instance Connect instead of SSH keys.)

Step 5: Configure Security Group

Create a new Security Group.

Name: CloudLabSG

Allow:

- SSH (Port 22)
- HTTP (Port 80)

Click Launch Instance.

Step 6: Connect to the Instance

After the instance status changes to Running:

1. Select the instance.
2. Click Connect.
3. Select EC2 Instance Connect.
4. Click Connect.

A terminal window opens.

Step 7: Update Ubuntu

Execute:

sudo apt update

Step 8: Install Apache

sudo apt install apache2 -y

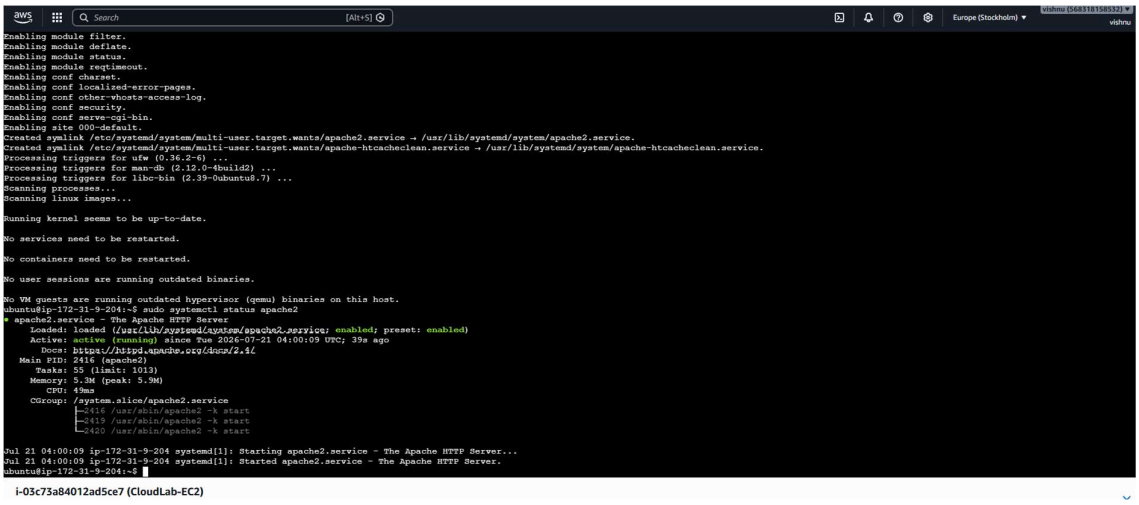
Verify installation:

sudo systemctl status apache2

The status should display:

active (running)

Attach a screenshot below:



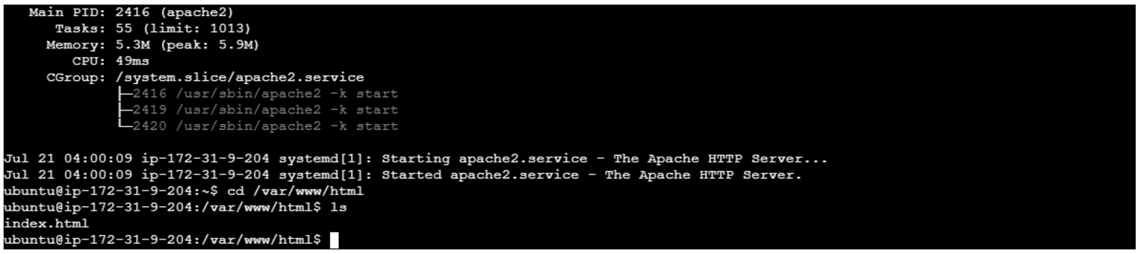
Step 9: Navigate to the Web Directory

cd /var/www/html

Display the files:

ls

Attach a screenshot below:



You should see:

index.html

Step 10: Replace the Default Web Page

Open the file:

sudo nano index.html -→ Now change the data in your static page

Step 11: Restart Apache

sudo systemctl restart apache2

Step 12: Obtain the Public IPv4 Address

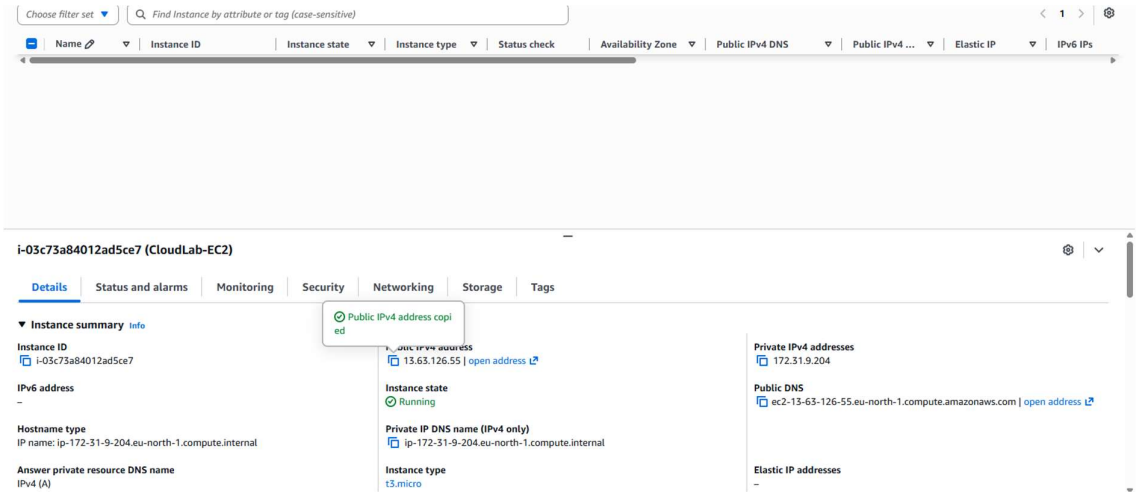
Return to the EC2 Dashboard.

Copy the Public IPv4 Address.

Example:

18.xxx.xxx.xxx

Attach a screenshot below:



Step 13: Access the Website

Open a browser.

Enter:

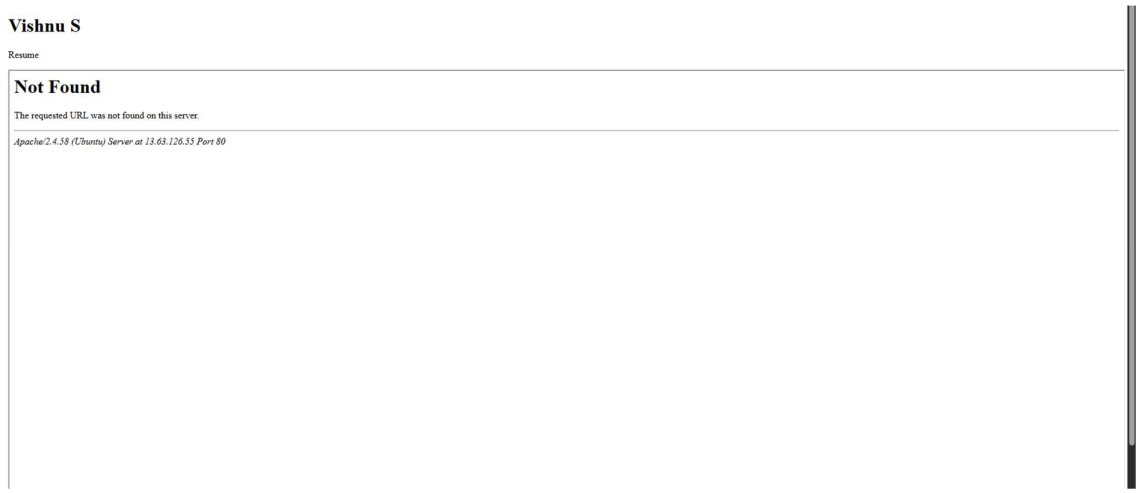
http://<Public-IP>

Example:

http://18.xxx.xxx.xxx

The custom webpage should be displayed.

Attach a screenshot below:



Step 14: Stop and Terminate the Instance

After completing the experiment:

1. Select the instance.
2. Click Instance State.
3. Choose Stop Instance.
4. After verification, choose Terminate Instance.

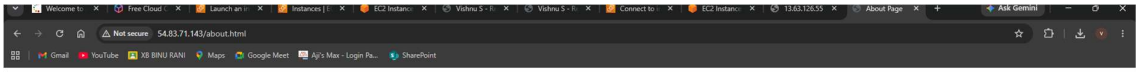
Important: Terminating the instance prevents unnecessary AWS charge

Task2:

Create a second page named about.html and link it from the home page.

Attach a screenshot below:




Not secure
54.83.71.143/about.html
Gmail
YouTube
XB BINU RANI
Maps
Google Meet
Aji's Max - Login Pa...
SharePoint
About Page
Ask Gemini

I am fine.


29°C
Partly sunny
Search
ENG
IN
10:28
21-07-2026